Model Based Testing and Internet of Things (IOT)

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Abstract

The Internet of Things (IoT) is changing how devices and systems work together, but testing these systems is difficult. Traditional testing methods often fail with many devices, connections, and real-time actions. Model-Based Testing (MBT) offers a more innovative way to test IoT systems by using system models to create test cases automatically. This report reviews five recent research papers exploring how MBT improves testing in IoT environments. These papers cover different testing goals—finding security problems, checking how systems behave when failures happen, and improving automation. We analyze their strengths, weaknesses, tools used, and how effective they are. Our findings show that while MBT is a powerful method, challenges like complexity and lack of fully developed tools exist. The report also suggests future directions to make MBT more practical and scalable for real-world IoT systems.

Keywords: Model-Based Testing, Internet of Things, Structural Testing, Security Testing, Safety Analysis, Automation, Test Generation, Test Coverage

1 Introduction

The Internet of Things (IoT) rapidly transforms industries by connecting everyday objects to the Internet. This connectivity allows these objects to collect data, make decisions, and interact with one another. IoT is becoming integral to daily life, with applications ranging from smart homes to industrial systems. However, this rapid growth presents a significant challenge: how can we effectively test these systems to ensure they function reliably, safely, and securely?[4]

Traditional software testing methods often struggle to address the unique characteristics of IoT systems. These systems involve multiple layers, including physical devices, sensors, and cloud platforms, and they operate in real-time environments. As a result, testing becomes complex and time-consuming. One promising solution is Model-Based Testing (MBT), which allows engineers to automatically create tests from models describing how the system should behave. Researchers have recently explored different ways to combine MBT with IoT testing. Some have focused on security testing, where the goal is to find vulnerabilities; others have worked on safety analysis, helping detect how faults might spread through a system before it is even built.[2] Some use automata learning to test systems without knowing their internal design, while others build tools that generate test models based on system rules and goals.[5]

This report reviews five key research papers that contribute to this area. These papers present different testing approaches, tools, and techniques to improve how we test IoT systems using MBT. By summarizing and analyzing these works, we aim to understand the current state of MBT for IoT, highlight what works well, what challenges still exist, and suggest where future research should go.

2 Justification of Selected Area

The rapid growth of the Internet of Things (IoT) has led to the emergence of interconnected systems across sectors like healthcare, smart cities, and automotive industries. Due to their real-time and safety-critical nature, ensuring their correctness, reliability, and security is essential. Traditional testing methods often struggle with the complexity of IoT systems. Model-Based Testing (MBT) offers a practical solution by enabling automated test generation from abstract models, improving test coverage and fault detection. Integrating MBT into IoT systems provides a structured way to tackle various testing challenges, making it vital for their robust development and deployment.

3 Research Objectives and Research Questions

This report addresses the following research objectives and research questions to better understand the intersection of MBT and IoT testing:

3.1 Research Objectives

The main objectives of this report are:

- To explore how Model-Based Testing (MBT) techniques are being applied in the context of Internet of Things (IoT) systems.
- To analyze the current methodologies, tools, and frameworks proposed in the research literature for testing IoT systems using MBT.
- To identify the strengths, limitations, and gaps in the existing MBT approaches for IoT testing.
- To compare the contributions of different research papers in terms of automation, fault detection, model design, and tool support.
- To suggest possible future directions for improving MBT practices in IoT environments.

3.2 Research Questions

Based on the above objectives and the literature reviewed, this report aims to answer the following research questions:

- **RQ1:** How is Model-Based Testing currently applied in the testing of IoT systems across different domains?
- **RQ2:** What types of models, tools, and formalisms are used to support MBT in IoT environments?
- **RQ3:** What are the common challenges and limitations identified in recent research on MBT for IoT?
- **RQ4:** How effective are MBT approaches in improving test coverage, automation, and reliability in IoT systems?
- **RQ5:** What future directions are proposed in the literature to enhance the scalability, usability, and integration of MBT in IoT?

4 Summary of the papers

A Systematic Review of IoT Systems Testing: Objectives, Approaches, Tools, and Challenges

Citation: Minani, Jean Baptiste, Fatima Sabir, Naouel Moha, and Yann-Gaël Guéhéneuc. *IEEE Transactions on Software Engineering* 50, no. 4 (2024): 785–815.

This systematic literature review examines the current approaches to testing in Internet of Things (IoT) systems, concentrating on testing goals, methodologies, tools, and challenges.The motivation for this review stems from the growing complexity of Internet of Things (IoT) environments, which include devices, networks, cloud services, and applications. This complexity has made traditional software testing techniques insufficient. Because of their unique structures and behaviors, IoT systems demand more tailored, scalable, and real-time testing strategies.The authors aim to consolidate fragmented research and provide a clear overview of IoT testing across various domains. Employing the PRISMA framework, they screened over 8,000 articles and selected 83 relevant studies published between 2012 and 2022. These studies were analyzed on the basis of testing goals (such as functionality, security, scalability, and energy efficiency), techniques (including model-based testing and fault injection), and the available tools and testbeds.

The findings indicate that most of the testing occurs at the integration and system levels. Although several tools are available, from prototypes to open source platforms, few are mature enough for real-world deployment or support DevOps practices such as continuous integration. Testbeds are widely used; however, the datasets they employ are often private and lack reproducibility. Major challenges include the absence of standardized evaluation frameworks and the difficulties associated with testing in highly dynamic, multi-layered environments. This study is notable for its comprehensive scope and structured analysis. Although it does not address some areas, such as emulator-based testing and security-specific testing, it effectively maps the current landscape and identifies key gaps. The authors conclude by calling for more unified testing frameworks, open benchmarking datasets, and robust tools that accurately reflect the realities of IoT development and deployment.[4]

Supporting Early-Safety Analysis of IoT Systems by Exploiting Testing Techniques

Citation: Clerissi, D., Di Rocco, J., Di Ruscio, D., Di Sipio, C., Ihirwe, F., Mariani, L., ... & Rubei, R. (2023, October). Supporting early-safety analysis of IoT systems by exploiting testing techniques. In *2023 ACM/IEEE International Conference on Model Driven Engineering Languages and Systems Companion (MODELS-C)* (pp. 520–529). IEEE.

This paper presents a model-driven safety assurance approach for IoT systems that enhances the completeness and accuracy of Failure Logic Analysis (FLA) by integrating model-based testing (MBT). The study is motivated by the unique challenges posed by IoT systems, where failures, whether triggered internally or by the surrounding environment, can spread across components and significantly impact system reliability and safety. Current FLA approaches are heavily based on manually specified failure rules, which are often incomplete or incorrect. The objective here is to use systematic testing to improve the identification and refinement of failure propagation rules early in the development lifecycle. The proposed methodology combines architectural modeling using CHESSIoT (a UML/SysML-based profile for IoT systems) with failure injection and test execution. The process consists of three phases: (1) IoT system modeling, where components and their failure behavior are specified; (2) fault tree generation and analysis, which identifies potential paths of failure propagation; and (3) IoT system testing, where test cases are executed with injected failures to observe propagation and validate or refine the FLA rules. The approach is demonstrated using a Smart Irrigation System case study representing a real-world IoT application with sensors, actuators, and cloud services. The preliminary evaluation shows that integrating test results into fault tree analysis (FTA) significantly improves early-stage safety assessments. Key strengths include practical application and potential automation of model-based rule refinement. However, a notable limitation is the lack of validation on larger or more diverse Internet of Things (IoT) systems. This work is distinctive for its test-informed feedback loop, offering a semi-automated approach to ensure safety modeling accuracy. It expands model-based testing (MBT) into the largely unexplored area of early safety verification in IoT.

Evaluation is preliminary but offers promising insights, indicating that integrating test results into fault tree analysis (FTA) significantly improves the accuracy of early-stage safety assessments. This work has several strengths, including its practical application and the potential for automating model-based rule refinement. However, a major limitation is the lack of validation in larger or more diverse Internet of Things (IoT) systems. In comparison to the existing FTA literature, this work is unique due to its test-informed feedback loop, which provides a semi-automated method to ensure correctness and completeness in safety modeling. It advances model-based testing (MBT) by exploring early safety verification, an area that remains underexplored in the IoT.[1]

Model-Based Security Testing in IoT Systems: A Rapid Review

Citation: Lonetti, F., Bertolino, A., & Di Giandomenico, F. (2023). Model-based security testing in IoT systems: A Rapid Review. *Information and Software Technology*, 164, 107326.

This paper explores how model-based security testing (MBST) is used to tackle the growing security challenges in Internet of Things (IoT) systems. The motivation behind this study stems from the increasing complexity and vulnerability of IoT environments. With billions of interconnected devices—from smart homes and healthcare systems to critical infrastructures—the attack surface is broader than ever, yet traditional testing methods are falling short. MBST, which generates test cases from formal models of the system, offers a promising path toward more structured, automated, and effective security validation in IoT. The main objective of this rapid review is to systematically analyze and classify existing research on MBST in the IoT domain. While MBT has been widely studied in general software systems, its application to security testing in IoT has not yet been thoroughly examined. This paper seeks to fill that gap by addressing questions around the testing objectives, model formalisms, test generation techniques, targeted domains, and types of attacks considered in MBST for IoT.

The authors used a Rapid Review methodology to analyze 803 publications retrieved from Scopus and Google Scholar. After applying clear inclusion/exclusion criteria and a rigorous snowballing process, they narrowed the list to 17 primary studies published over the last five years. These studies are then categorized using various analytical dimensions: types of models, testing goals, targeted attacks, application domains, tools used, and so on. The key findings reveal that UML models (often with OCL constraints) and timed automata are the most frequently used formalisms. Test case generation typically relies on structural coverage of models, test purposes, or model-checking tools like UPPAAL. The most targeted attacks include Denial of Service (DoS), dictionary attacks, eavesdropping, and data leakage. The most commonly studied domains are application-wise, automotive systems, and smart city infrastructures. MBST is also used in healthcare, smart homes, and generic IoT protocol validation. Tools like CertifyIt, TTCN-3, and TITAN are often used for test generation and execution. This paper's main contribution is its first-ever systematic classification of MBST literature in the IoT space. It offers a detailed mapping of testing approaches and highlights how MBST can be combined with other techniques like fuzzing, penetration testing, and threat modeling for a more comprehensive defense. In terms of evaluation, although the review does not assess the effectiveness of each method quantitatively, it offers a thorough qualitative synthesis. The structured approach ensures reliability, and including peer-reviewed, recent work adds to its relevance.

A notable strength of the paper is that it not only categorizes existing research but also identifies gaps and future directions, such as the need for dynamic modeling techniques, better support for evolving IoT architectures, and integration with AI-driven testing tools. A limitation, however, is the relatively small number of primary studies, which reflects how early the field still .Compared to previous reviews that focused on MBT in general or IoT testing more broadly, this work is distinct in focusing exclusively on MBST within the IoT context. It offers a foundation for future researchers and practitioners to develop robust, scalable, and adaptive security testing frameworks tailored to IoT ecosystems.[2]

Model-Based Testing IoT Communication via Active Automata Learning

Citation: Tappler, M., Aichernig, B. K., & Bloem, R. (2017, March). Model-based testing IoT communication via active automata learning. In *2017 IEEE International Conference on Software Testing, Verification and Validation (ICST)* (pp. 276–287). IEEE.

This study proposes a novel application of model-based testing (MBT) through active automat learning to detect faults in communication protocols used in IoT systems, focusing on MQTT. The research is motivated by the observation that communication failures among heterogeneous IoT systems are hard to detect due to the absence of shared structural models and the limitations of conventional test generation. This work aims to present a semi-automated method for discovering protocol violations across different implementations of MQTT brokers, components central to publish / subscribe-based IoT communication. The core methodology involves learning Mealy machine models from black-box implementations of MQTT brokers using the LearnLib tool and then pairwise cross-checking these models to detect deviations. When two brokers exhibit different behaviors for identical input sequences, these discrepancies are flagged as potential bugs. The case study includes five MQTT implementations, revealing faults in four out of five, demonstrating the practical value of this approach. The method does not require prior modeling of the system under test, making it adaptable for testing third-party or legacy systems. Evaluation is carried out through hands-on experiments that involve real-world IoT communication protocols, offering empirical foundation. Strengths include the ability of the method to uncover hidden flaws without access to source code, which is especially valuable for proprietary or undocumented implementations. Limitations involve the semiautomatic nature of the process, as manual effort is still needed to analyze counterexamples and the inability to detect faults common across all implementations. This study stands out from previous MBT literature by eliminating the need for hand-crafted models, instead using inferred models to perform conformance checking. It opens the door to scalable learning-driven MBT solutions in IoT communication.[5]

A Multimethod Study of Internet of Things Systems Testing in Industry

Citation: Minani, J. B., Sabir, F., Moha, N., & Guéhéneuc, Y. G. (2023). A multimethod study of Internet of Things systems testing in industry. *IEEE Internet of Things Journal*, 11(1), 1662–1684.

This multimethod empirical study investigates how Internet of Things (IoT) systems are tested in real-world industrial environments to bridge the gap between academic research and industrial practice. The motivation for this study arises from a pressing concern: while IoT systems have become integral to critical domains such as healthcare, transportation, and infrastructure, most proposed testing strategies originate from academia and have limited validation in real industry settings. As IoT continues to increase in complexity, ensuring system dependability through testing has become increasingly vital—but also uniquely challenging due to the multi-layered architecture of IoT systems, their heterogeneous components, and the lack of standardized testing practices. The authors aim to explore how IoT testing is conducted in the industry, focusing on tools, testing approaches, quality attributes, test automation, and the significant challenges practitioners face. Their objectives are clear: (1) identify the testing tools and techniques used by practitioners, (2) determine which quality attributes are prioritized, (3) discover which test artifacts are commonly automated, and (4) understand the pain points developers encounter during testing. To achieve this, the authors conduct a multimethod study combining an online industry survey (49 participants), semi-structured interviews (9 professionals), and a longitudinal analysis of Eclipse IoT developer surveys from 2018 to 2022.

Key findings indicate that integration testing is the most prevalent practice, while acceptance testing receives less emphasis. The application, network, and device layers are tested most frequently, whereas testing in the cloud and business layers is rare. Testing remains primarily manual or semi-automated, with Node-RED identified as the most widely used tool, followed by AWS IoT for cloud-level testing. Interestingly, although nearly half of the respondents acknowledged model-based testing (MBT), white-box testing remains uncommon. For root cause analysis, practitioners predominantly rely on manual log analysis, highlighting the lack of specialized debugging tools. Performance, security, connectivity, and interoperability are deemed essential quality attributes; however, few participants employ formal metrics or systematic test coverage approaches, often relying on project-specific criteria instead. The contribution of this study lies in its grounded, practitioner-centered perspective. Map actual testing practices, list tools and methods, and classifies the industry's challenges. These challenges include a lack of test standards and reference architectures, issues with test case generation, and difficulties in managing protocol heterogeneity. Future research should focus on automated test case generation, developing IoT-specific standards, and improving tool support for testing across different layers.

In terms of evaluation and validation, this study stands out for its methodological rigor—it applies the SPIDER framework, conducts interviews to complement survey data, and draws upon a broad longitudinal dataset from Eclipse IoT. The strength of this work is its practical relevance and empirical depth, providing a critical industry perspective on IoT testing. Limitation, however, is that while it captures a wide range of testing experiences, it does not deeply evaluate the effectiveness of individual tools or techniques. Compared to existing literature, this paper is unique in being industry-centric, offering real-world insights that contrast with the theory-heavy, lab-based focus of prior research.[3]

Paper Title	Strengths	Limitations
A Systematic Review of IoT Systems Testing	Covers 83 studies comprehensively; provides structured analysis of goals and challenges; identifies gaps such as lack of standardized frameworks	Does not address emulator-based testing; lacks security-specific testing focus
Supporting Early-Safety Analysis of IoT Systems	Includes real-world Smart Irrigation case study; provides semi-automated rule refinement; enhances fault tree analysis accuracy	Limited scalability validation; model refinement still partially manual
Model-Based Security Testing in IoT Systems	First structured review of MBST in IoT; identifies modeling gaps; proposes integration with AI tools	Only 17 studies reviewed; provides qualitative not quantitative synthesis
Model-Based Testing IoT Communication	Detects protocol flaws without code access; supports scalable black-box testing; avoids handcrafted models	Requires manual analysis of counterexamples; fails to detect universal implementation faults
A Multimethod Study of IoT Testing in Industry	Offers industry-focused insights via surveys/interviews; identifies practical testing issues; connects academic and industrial perspectives	Tool effectiveness not deeply evaluated; limited generalizability due to small sample

Table 1: Comparison of Strengths and Limitations in Reviewed Papers

5 Conclusion

This report examines how Model-Based Testing (MBT) improves testing for Internet of Things (IoT) systems. After reviewing five research papers, we found that MBT offers key benefits like better test coverage, a more organized approach, and support for automation. It helps tackle IoT challenges, such as security, early failure detection, and communication protocol testing. However, many tools are still prototypes and not ready for large-scale use. Managing the complexity of IoT systems remains challenging due to their dynamic nature. Overall, MBT is a promising method, but it needs to become more practical and user-friendly. Future efforts should focus on developing reliable tools and improving integration with development processes

References

[1] Diego Clerissi, Juri Di Rocco, Davide Di Ruscio, Claudio Di Sipio, Felicien Ihirwe, Leonardo Mariani, Daniela Micucci, Maria Teresa Rossi, and Riccardo Rubei. Supporting early-safety analysis of iot systems by exploiting testing techniques. In *2023 ACM/IEEE International Conference on Model Driven Engineering Languages and Systems Companion (MODELS-C)*, pages 520–529. IEEE, 2023.

[2] Francesca Lonetti, Antonia Bertolino, and Felicita Di Giandomenico. Model-based security testing in iot systems: A rapid review. *Information and Software Technology*, 164:107326, 2023.

[3] Jean Baptiste Minani, Fatima Sabir, Naouel Moha, and Yann-Gaël Guéhéneuc. A multimethod study of internet of things systems testing in industry. *IEEE Internet of Things Journal*, 11(1):1662–1684, 2023.

[4] Jean Baptiste Minani, Fatima Sabir, Naouel Moha, and Yann-Gaël Guéhéneuc. A systematic review of iot systems testing: Objectives, approaches, tools, and challenges. *IEEE Transactions on Software Engineering*, 50(4):785–815, 2024.

[5] Martin Tappler, Bernhard K. Aichernig, and Roderick Bloem. Model-based testing iot communication via active automata learning. In *10th IEEE International Conference on Software Testing, Verification and Validation (ICST)*, pages 276–283. IEEE, 2017.